

03 - Baseline Risk

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

Before asking how much a treatment helps, ask how much risk is present before treatment.

This is not a statistical courtesy. It is bedside medicine. A relative risk reduction has different meaning when the baseline risk is three percent, sixteen percent, thirty-four percent, or fifty-seven percent. The patient's history and cervical length determine the probability space in which treatment is being considered.

Cervical-Length Risk

Start again with cervical length.

The Iams study measured cervical length around twenty-four weeks and showed a graded relationship with spontaneous preterm delivery before thirty-five weeks. The risk did not suddenly appear below a magic line. It rose as the cervix shortened.

At or below twenty-six millimeters, roughly the tenth percentile, the relative risk was about six point two compared with women above the seventy-fifth percentile. At or below twenty-two millimeters, the relative risk was about nine point five. At or below thirteen millimeters, it was about fourteen.

The memory is the gradient. Twenty-six, twenty-two, thirteen: shorter cervix, higher risk. The numbers are not meant to be recited as an isolated chant. They are meant to make the cervix-as-continuum model unforgettable.

Absolute Risk

Now translate that gradient into absolute risk. In the Iams cohort, a cervix of twenty-five millimeters or less at about twenty-four weeks corresponded to about an eighteen percent probability of spontaneous delivery before thirty-five weeks. A cervix of thirty millimeters or less corresponded to about nine percent. Twenty millimeters or less was higher, about twenty-six percent. So if someone says that the risk with a short cervix is around seven to fifteen percent, the next thought should be: which cervix, which population, and which endpoint?

Society for Maternal-Fetal Medicine gives a useful contemporary nulliparous screening frame. In a cohort of more than nine thousand nulliparas, cervical length twenty-five millimeters or less at sixteen to twenty-two weeks had a positive predictive value of about sixteen point two percent for spontaneous preterm birth

before thirty-seven weeks, but only about seven point four percent for spontaneous preterm birth before thirty-two weeks. The same cervix threshold gives different risks because the endpoint changed. Before thirty-seven and before thirty-two are not the same question.

Trial Placebo Risk

That continuum also explains why trial placebo arms differ.

In the Fonseca trial, women were selected from routine mid-trimester cervical screening, and the cervix had to be fifteen millimeters or less. That is a very short cervix. In the placebo group, spontaneous delivery before thirty-four weeks occurred in thirty-four point four percent. Store that as a baseline-risk number. It is not a treatment result. It tells you how enriched the population was for early preterm birth.

In the PREGNANT trial, the population was different. These were asymptomatic singleton pregnancies with cervical length between ten and twenty millimeters at nineteen to twenty-three plus six weeks. The placebo risk of preterm birth before thirty-three weeks was sixteen point one percent.

Do not force these two numbers to be the same. Fonseca used fifteen millimeters or less and an under-thirty-four endpoint. PREGNANT used ten to twenty millimeters and an under-thirty-three endpoint. Different entry criteria and different endpoint, different baseline risk.

This is why the phrase "short cervix has a risk of early labor of seven point five to fifteen percent" is not a satisfying evidence statement. Risk of what endpoint? In what population? With what cervical length? At what gestational age? With or without prior spontaneous preterm birth? A number without its denominator and endpoint is not clinical knowledge.

Recurrence History

Now add obstetric history.

McManemy studied recurrence risk in women with three consecutive singleton live births. The key result is beautifully clinical because it shows how history behaves like a probability engine.

If the first two births were term, the risk of preterm delivery in the third birth was about five percent. If the first birth was preterm and the second was term, the risk was thirteen percent. If the first was term and the second was preterm, the risk was twenty-one percent. If both prior births were preterm, the risk was forty-two percent.

The order matters. A more recent preterm birth carried more risk than a more remote one. The number of events matters. Two prior preterm births carried much more risk

than one. Severity matters too. Among women with two prior very preterm deliveries, between twenty-one and thirty-one weeks, the recurrence risk reached fifty-seven percent.

This is not just a list of percentages. It is a structure.

Five percent means no prior preterm birth in that sequence. Thirteen and twenty-one percent mean one prior preterm birth, but order changes the risk. Forty-two percent means two prior preterm births. Fifty-seven percent means two prior very preterm births. The memory is frequency, order, severity.

History Plus Current Biology

Now connect history and cervical length.

A twenty-two millimeter cervix in a patient with no prior preterm birth is already meaningful. The Iams gradient tells you it is not benign. But a twenty-two millimeter cervix in a patient with two prior very early spontaneous preterm births lives in a much more dangerous probability space. The ultrasound number did not change. The prior probability changed.

The Preterm Prediction Study made this concrete in parous women. Among women with prior preterm birth and a positive fetal fibronectin, the estimated risk of spontaneous preterm birth before thirty-five weeks was about sixty-five percent when the cervix was twenty-five millimeters or less, forty-five percent when the cervix was twenty-six to thirty-five millimeters, and twenty-five percent when the cervix was more than thirty-five millimeters. If fetal fibronectin was negative, the same cervical-length strata were about twenty-five percent, fourteen percent, and seven percent.

That matrix should permanently change how you hear the phrase "history of preterm birth." Positive fibronectin plus a short cervix is not the same clinical state as negative fibronectin plus a long cervix. Both patients have history. They do not have the same current probability.

That is why baseline risk must be taught before treatment effect. If a trial reduces a relative risk by about forty percent, the absolute benefit depends on where the patient started. A high-risk patient can have a larger absolute benefit. A low-risk patient may have a smaller absolute benefit, and a screening program in a low-prevalence population may show limited public-health effect even when a treatment is biologically useful in selected individuals.

This also protects counseling.

A patient should not hear a generic statement that "the risk is high." She should hear a probability that belongs to her phenotype. If she has a very short cervix before twenty-four weeks and no prior preterm birth, we should explain the short-cervix trial evidence. If she has a prior spontaneous preterm birth, we should explain recurrence

risk and how cervical length modifies it. If she presents after twenty-four weeks, we should explain that late detection changes the evidence.

The resident should leave this module with one reflex: never quote a risk number naked.

Attach it to the source population. Attach it to the endpoint. Attach it to gestational age. Attach it to cervical length. Attach it to prior history.

Only then is the number useful enough to guide treatment.